

CSE 815: Machine Learning

Week 1, Part 1 — Introduction to Machine Learning

ROKAN UDDIN FARUQUI

Professor

Department of Computer Science & Engineering

University of Chittagong, Chittagong

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Course & Lecture Information

Course Details

Code: CSE 815
Credits: 3 (Theory)
Pre-req: CSE 221 (Data Structures),
CSE 225 (Algorithms),
CSE 235 (Probability & Stats)

Today

Week / Part: 1 of 13 | Part 1 of 2
CLO: CLO1
PLO: PLO(a)

CLO1

“Understand the fundamental concepts, paradigms, and theoretical foundations of machine learning”

Part 1 Covers

- What is Machine Learning?
- The formal definition (Mitchell, 1997)
- The three learning paradigms
- The ML pipeline
- Key concepts: generalization, overfitting, underfitting, inductive bias
- Ivy League alignment
- Student assessment pool

Lecture Agenda

- 1 The Machine Learning Revolution
- 2 Formal Definition (Mitchell, 1997)
- 3 The Three Learning Paradigms
- 4 The Machine Learning Pipeline
- 5 Key Concepts: Overfitting, Underfitting, Generalization
- 6 Ivy League Alignment
- 7 Student Assessment Pool

The Machine Learning Revolution

Why Machine Learning Now?

The AI Transformation

Machine learning has moved from academic research to **every aspect of modern life**:

- **Consumer**: Netflix recommendations, Amazon product suggestions, Spotify Discover Weekly
- **Communication**: Gmail spam filter, Google Translate, voice assistants (Siri, Alexa)
- **Transportation**: Tesla Autopilot, Uber surge pricing, Waze routing
- **Healthcare**: Disease diagnosis from medical imaging, drug discovery (AlphaFold)
- **Finance**: Fraud detection, credit scoring, algorithmic trading

Market Context

Global AI market: projected **\$1.8 trillion by 2030** (industry forecasts)
ML/AI roles: among the **fastest-growing job categories** worldwide (WEF *Future of Jobs*)
Bangladesh: AI/ML a **fast-growing segment** of national ICT exports

Why Now? (4 Drivers)

- **Data explosion**: global datasphere $\approx$ 175 ZB (IDC projection, 2025)
- **Compute power**: GPUs, TPUs, cloud
- **Algorithms**: Deep learning breakthroughs
- **Open source**: TensorFlow, PyTorch, scikit-learn democratised access

Timeline of AI & ML Milestones

1959	Samuel – self-improving checkers player
1986	Backpropagation popularised (Rumelhart et al.)
1997	Deep Blue beats Kasparov (<i>search, not learning</i>)
2012	AlexNet – ImageNet breakthrough
2016	AlphaGo beats Lee Sedol
2020	AlphaFold solves protein structure prediction
2022	ChatGPT – LLMs reach the public
2024	Nobel Prizes recognise ML research

What Makes ML Different from Traditional Programming?

Traditional Programming

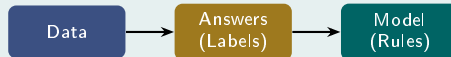


Human writes rules; machine executes them.

Example: Spam filter with hand-coded rules

- IF sender NOT in contacts THEN flag
- IF contains "lottery" THEN flag
- (1000+ rules maintained by engineers)

Machine Learning



Machine learns rules from data + answers. **Example:**

ML spam filter

- Train on 100,000 labeled emails
- Model learns patterns automatically
- No human-written rules needed
- Achieves 99.9% accuracy

The Paradigm Shift

Traditional programming: **explicit** knowledge encoding.

Machine learning: **implicit** knowledge discovery from data.

Formal Definition (Mitchell, 1997)

The Canonical Definition: Tom Mitchell (1997)

Full Citation

Mitchell, T. M. (1997). **Machine Learning**. McGraw-Hill. *The first comprehensive textbook; still a standard reference.*

The Definition

*“A computer program is said to **learn** from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, **improves with experience E.**”* —Tom Mitchell, 1997

Deconstructing the Definition: Spam Filter

- **Task T**: Classify emails as spam or not spam
- **Performance P**: Accuracy (% correctly classified)
- **Experience E**: 100,000 labeled emails (spam/ham)

Performance improves as the model sees more labeled emails.

Why This Definition Matters

- Provides a **rigorous framework** for what “learning” means
- Every ML problem can be specified as **T, P, E**
- Forces practitioners to be **precise** about the task, the success metric, and the available data
- Enables **comparison** across algorithms

The Earlier, Informal Definition

“The field of study that gives computers the ability to learn without being explicitly programmed.”
—attributed to Arthur Samuel, 1959

Widely quoted, but a later paraphrase—it does not appear verbatim in Samuel’s 1959 paper (examined in Part 2).

Applying Mitchell's Framework: Examples

Problem	Task (T)	Experience (E)	Performance (P)
Spam detection	Classify email as spam/ham	1M labeled emails	Accuracy, F1-score
House price prediction	Predict price (regression)	Historical sales records	RMSE, R^2
Credit scoring	Predict default risk	Loan repayment history	ROC-AUC
Product recommendation	Rank products for a user	User purchase history	Click-through rate
Image classification	Identify object in image	Labeled image dataset	Top-1 / Top-5 accuracy
Fraud detection	Flag fraudulent transaction	Past labeled transactions	Precision / Recall

The Critical Insight

Without a clearly defined T, P, E, you are not doing machine learning—you are just writing code. The definition keeps us honest about the goal.

The Three Learning Paradigms

Supervised Learning: Learning with Labeled Data

Definition

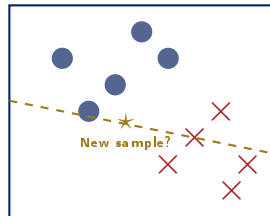
Supervised learning learns a mapping $f : x \mapsto y$ from labeled examples $\{(x_i, y_i)\}_{i=1}^n$, then predicts y for unseen x .

Two Main Sub-types

- **Classification:** discrete class label
 - Binary (spam vs. ham), multi-class (digits 0–9), multi-label (image tags)
- **Regression:** continuous value
 - House price, temperature, demand forecasting

Local Example

Mobile-money credit scoring (e.g. bKash): predict loan default risk—binary classification trained on past transactions with known repayment outcomes.



Training Data (labeled)

Key Algorithms

Linear Regression, Logistic Regression, SVM, Decision Trees, Random Forest, Neural Networks

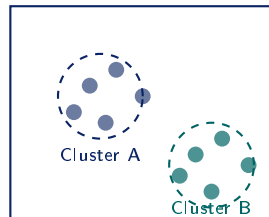
Unsupervised Learning: Finding Hidden Structure

Definition

Unsupervised learning discovers hidden structure in **unlabeled** data $\{x_i\}_{i=1}^n$ —no target values are provided.

Main Sub-types

- **Clustering**: Group similar data points together
 - Customer segmentation
 - Image compression
 - Document organisation
- **Dimensionality Reduction**: Find lower-dimensional representation
 - PCA, t-SNE, UMAP
 - Data visualization
 - Feature extraction
- **Anomaly Detection**: Identify unusual data points
 - Fraud detection, equipment failure prediction



Unlabeled Data

Key Algorithms

K-Means, Hierarchical Clustering, DBSCAN, PCA, t-SNE

Reinforcement Learning: Learning from Actions & Rewards

Definition

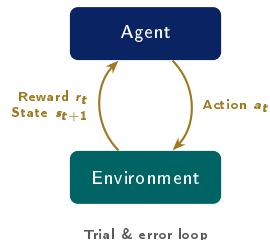
Reinforcement learning learns a **policy** by **trial and error**, maximising cumulative **reward**. An agent acts in an **environment** and receives rewards or penalties.

Key Components

- **Agent**: the learner / decision-maker
- **Environment**: what the agent acts upon
- **State**: current situation; **Action**: what it may do
- **Reward**: scalar feedback (positive or negative)
- **Policy**: strategy mapping states to actions

Game Playing (AlphaGo)

State = board position; Actions = possible moves; Reward = win/loss.
AlphaGo learned by playing millions of games against itself.



Key Applications

- Game playing (AlphaGo, chess, poker)
- Robotics, control, autonomous vehicles
- Resource management, sequential recommendation

The Three Paradigms: Comparison

Dimension	Supervised	Unsupervised	Reinforcement
Data	Labeled (x, y) pairs	Unlabeled x only	Sequential states, actions, rewards
Goal	Predict y from x	Discover hidden structure	Maximize cumulative reward
Feedback	Immediate (label known)	None (no ground truth)	Delayed and evaluative
Key challenge	Generalization	Evaluation / interpretability	Credit assignment; exploration vs. exploitation
Example	Spam filter	Customer segmentation	AlphaGo

Beyond the Three

Semi-supervised: a few labels + much unlabeled data.

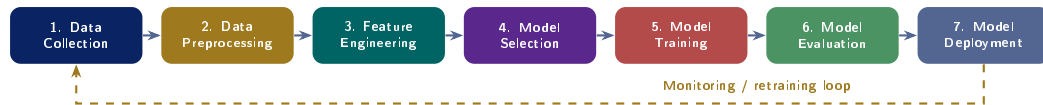
Self-supervised: labels generated from the data itself (masked-token prediction)—the engine behind modern LLMs.

Course Focus

CSE 815 covers **supervised and unsupervised learning** in depth (Weeks 3–11). Reinforcement learning is treated **conceptually only**—it is defined and contrasted here, but not developed or assessed algorithmically.

The Machine Learning Pipeline

The End-to-End ML Pipeline



Stage	Description
1. Data collection	Gather raw data from sources (databases, APIs, sensors, surveys)
2. Data preprocessing	Clean data: handle missing values, remove duplicates, scale/normalise
3. Feature engineering	Transform raw data into features the model can use
4. Model selection	Choose algorithm(s) and tune hyperparameters via validation data
5. Model training	Learn parameters from the training data
6. Model evaluation	Assess performance on held-out test data, once
7. Model deployment	Serve in production, monitor for data/concept drift , retrain

Key Insight

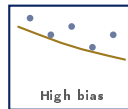
In real-world ML, **stages 1–3 (data work)** typically consume the majority of project time (practitioner surveys report $\sim 80\%$). The modelling steps are the smallest part—contrary to public perception. (Cf. *Sculley et al., 2015, “Hidden Technical Debt in ML Systems”, Week 12.*)

Key Concepts: Overfitting, Underfitting, Generalization

The Central Challenge: Generalization

Generalization

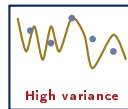
The ability of a model to perform well on **new, unseen data**—not merely on the data it was trained on. This, not training accuracy, is **the goal** of machine learning.



Underfitting



Just Right



Overfitting

How We Measure It: Data Splits

- **Training set** (~60–80%): fit model parameters
- **Validation set** (~10–20%): tune hyperparameters, select models
- **Test set** (~10–20%): **touched once**, for the final unbiased estimate

Generalization gap = test error – training error.

Model complexity → (left: too simple; right: too complex)

The Goldilocks Problem

A model must be **complex enough** to capture the pattern, yet **simple enough** to generalise. This tension is formalised next as the **bias–variance trade-off**.

Golden Rule

Never make a modelling decision using the test set—that leaks information and inflates reported performance.

Overfitting vs. Underfitting: A Closer Look

Dimension	Overfitting	Underfitting
Definition	Model learns noise, not signal	Model fails to capture the underlying pattern
Training error	Very low	High
Test error	High	High
Model complexity	Too high	Too low
Dominant term	Variance	Bias
Cause	Too many parameters, insufficient data	Model too simple, too few features
Diagnosis	Train error $\ll$ test error	Train error $\approx$ test error (both high)
Fix	Regularization, more data, simpler model	Add features, use a more expressive model

Real-World Example: Medical Diagnosis

Underfitting: A model that always predicts “no disease” (high bias) — misses actual cases.

Overfitting: A model that memorises specific patient records — fails with new patients.

Goal: A model that learns the *general* relationship between symptoms and disease.

Inductive Bias: The Assumptions We Make

Definition

Inductive bias is the set of assumptions a learning algorithm makes to generalise beyond the training data. Without bias, no learning is possible.

Examples by Algorithm

- **Linear regression:** Assumes linear relationship between features and target
- **KNN:** Assumes similar inputs have similar outputs (smoothness)
- **Decision trees:** Assumes hierarchical feature relevance
- **SVM:** Assumes large-margin separation
- **Neural networks:** Assumes composition of functions

No Free Lunch Theorem (Wolpert, 1996)

No single algorithm is best for all problems. The “right” algorithm depends on the problem’s underlying structure—which we encode as inductive bias.

The Bias–Variance Decomposition

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

(exact for squared-error loss; a useful intuition otherwise)

- **Bias:** Error from incorrect assumptions
- **Variance:** Error from sensitivity to training data
- **Noise:** Irreducible error from the problem itself

The trade-off:

- High bias → underfitting
- High variance → overfitting
- Need to balance both

Key Insight

Strong inductive bias → learns faster, may be wrong
Weak inductive bias → more flexible, needs more data

Ivy League Alignment

Stanford CS229: The Gold Standard

Course Philosophy (Andrew Ng)

"Machine learning is the science of getting computers to act without being explicitly programmed. In the past decade, machine learning has given us self-driving cars, practical speech recognition, effective web search, and a vastly improved understanding of the human genome."

— Stanford CS229 Course Description

CS229 Coverage:

- Supervised learning: linear regression, logistic regression, SVMs, neural networks
- Unsupervised learning: clustering, PCA, factor analysis
- Learning theory: bias-variance, regularization
- Reinforcement learning: MDPs, value iteration, policy gradients

CS229's First Lecture

Andrew Ng opens with the Mitchell definition, the spam filter example, and the three paradigms. Students then implement linear regression in the first homework.

CSE 815 Alignment

This course follows CS229's **structure and philosophy**:

- Week 1: Introduction & foundations
- Weeks 2-7: Supervised learning
- Weeks 8-9: Unsupervised learning
- Weeks 10-11: Neural networks & deep learning
- Weeks 12-13: Advanced topics & ethics

Also Aligned With:

- MIT 6.867: Statistical inference foundation
- CMU 10-701: PhD-level foundations
- UC Berkeley CS189: Mathematical rigour
- Harvard CS181: Applied perspective

Student Assessment Pool

Instructions

Select the single best answer. Answers are **bolded** for instructor reference; present without bold to students.

- Q1. Who provided the canonical formal definition of machine learning as “*a computer program is said to learn from experience E with respect to some class of tasks T and performance measure P ...*”?
- ☐ (a) Arthur Samuel (b) **Tom Mitchell** (c) Geoffrey Hinton (d) Yann LeCun
- Q2. Which paradigm of machine learning uses **unlabeled data** to discover hidden structure?
- ☐ (a) Supervised learning (b) **Unsupervised learning** (c) Reinforcement learning (d) Semi-supervised learning

Instructor Notes

Q1: Students commonly confuse Samuel (who coined the term) with Mitchell (who gave the formal definition).
Q2: Reinforcement learning also uses unlabeled data but through interaction; the key distinction is the presence of rewards.

- Q3. A model that performs well on training data but poorly on test data is suffering from:
- ☐ (a) Underfitting
 - ☒ (b) **Overfitting**
 - ☐ (c) High bias
 - ☐ (d) Data leakage
- Q4. Which of the following is an example of a **supervised learning task**?
- ☐ (a) Customer segmentation
 - ☐ (b) Anomaly detection
 - ☒ (c) **House price prediction**
 - ☐ (d) Market basket analysis

Instructor Notes

Q3: Overfitting is the classic case of high training accuracy + low test accuracy.

Q4: Customer segmentation, anomaly detection, and market basket analysis are unsupervised learning tasks (no labels available).

Instructions

State True or False. Award full marks only if the student provides a **one-sentence justification**.

Statement	Answer
1. Reinforcement learning requires labeled training examples.	False
2. The goal of machine learning is to perform well on unseen data (generalization).	True
3. Spam filtering is an example of unsupervised learning.	False
4. Inductive bias refers to the assumptions a learner makes in order to generalize.	True

Justification Model Answers

- 1: RL uses rewards, not labels; it learns through trial and error.
- 3: Spam filtering uses labeled emails (spam/ham), making it supervised learning.
- 4: Inductive bias is the set of assumptions that enables generalization beyond training data.

Instructions

Answer each question in **100–150 words**. Use specific terminology covered in class.

- Q1. Define machine learning using Tom Mitchell's formal definition. Explain each component (Task T, Experience E, Performance P) with a concrete example.
- Q2. Differentiate between **supervised**, **unsupervised**, and **reinforcement learning**. Provide one real-world application for each.
- Q3. What is the difference between **overfitting** and **underfitting**? How does each affect model generalization?

Instructions

Answer in **200–300 words**. Show step-by-step reasoning. Partial credit is awarded for correct reasoning even with incomplete conclusions.

A1. Problem Formulation (Healthcare):

A hospital wants to build a system to predict patient readmission risk within 30 days of discharge.

- Formulate this as a machine learning problem using Mitchell's **T-P-E** framework.
- What type of learning paradigm is this?
- What challenges might arise in this healthcare application?

A2. Bias-Variance Analysis:

Compare the inductive bias of a linear model vs. a decision tree.

- How does each learner's bias affect its ability to generalize?
- In what scenarios would each be preferred?
- How do you diagnose whether bias or variance is the dominant problem?

Textbook & Reading References

Primary Textbooks for Week 1

- **[Mitchell]** Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill. **Chapter 1**
- **[Bishop]** Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. **Chapter 1**
- **[Murphy]** Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press. **Chapter 1**
- **[Hastie]** Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer. **Chapters 1, 2**
- **[Shalev-Shwartz]** Shalev-Shwartz, S., & Ben-David, S. (2014). *Understanding Machine Learning*. Cambridge. **Chapters 1, 2**

Supplementary—Covered in Week 1 Part 2 (Research Articles)

- Samuel, A. L. (1959). *Some Studies in Machine Learning Using the Game of Checkers*. IBM Journal. **[PRIMARY]**
- Jordan, M. I., & Mitchell, T. M. (2015). *Machine Learning: Trends, Perspectives, and Prospects*. Science, 349(6245). **[SURVEY]**
- Wolpert, D. H. (1996). *The Lack of A Priori Distinctions Between Learning Algorithms*. Neural Computation. **[NO FREE LUNCH]**

Questions & Discussion

Week 1, Part 1 | CSE 815 Machine Learning

“Machine learning is the science of getting computers to act without being explicitly programmed.”

— Andrew Ng

Next Lecture: Week 1, Part 2

Research Articles: Samuel (1959) · Jordan & Mitchell (2015)

Case Study: Spam Filtering (Classic ML Success Story)